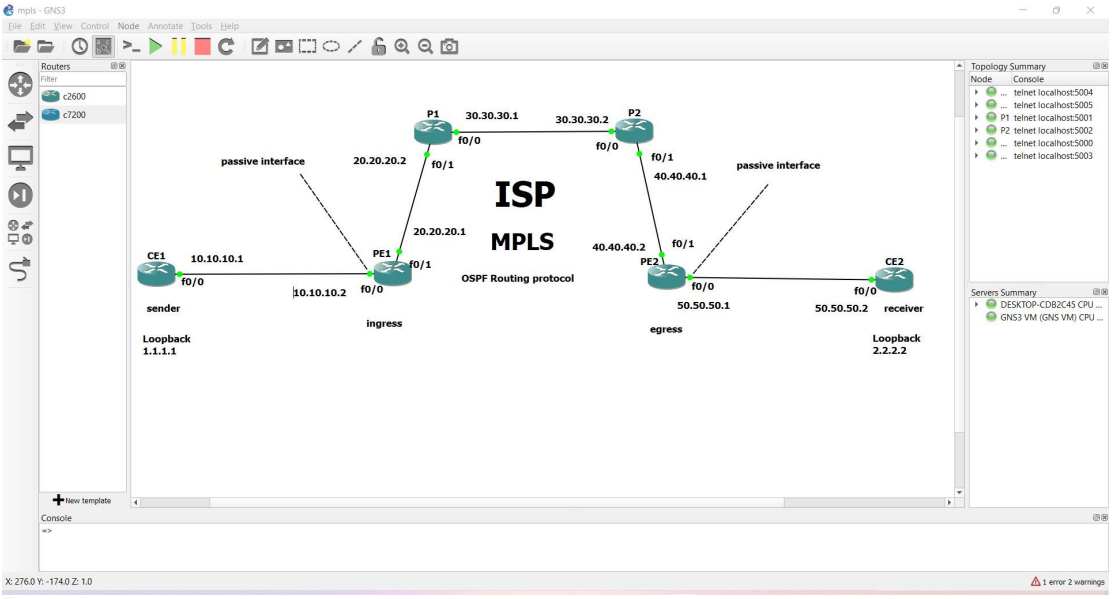
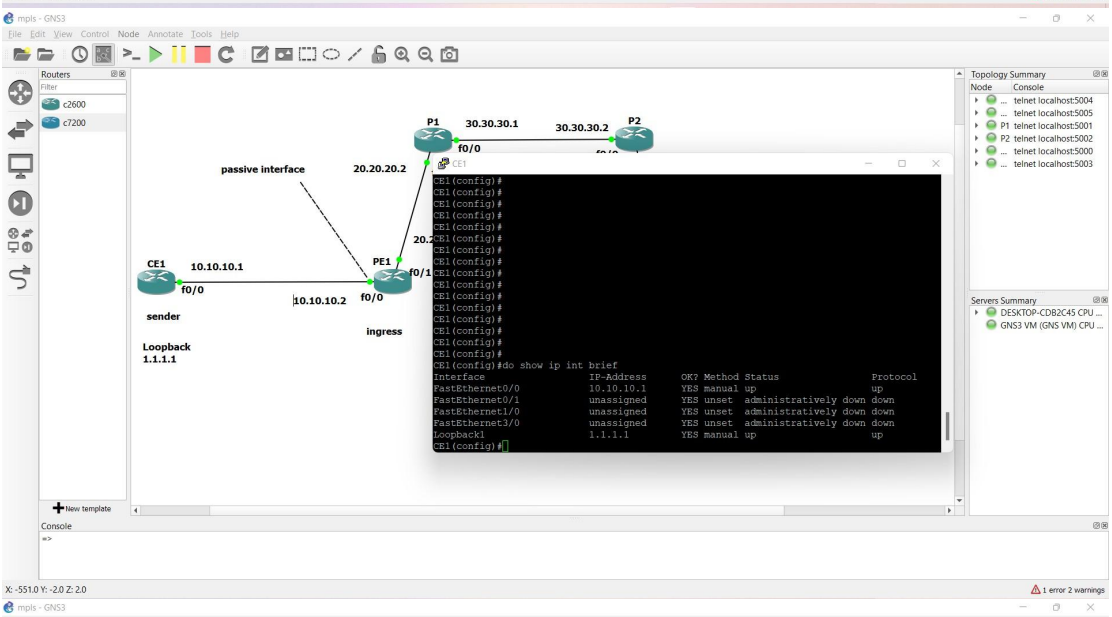
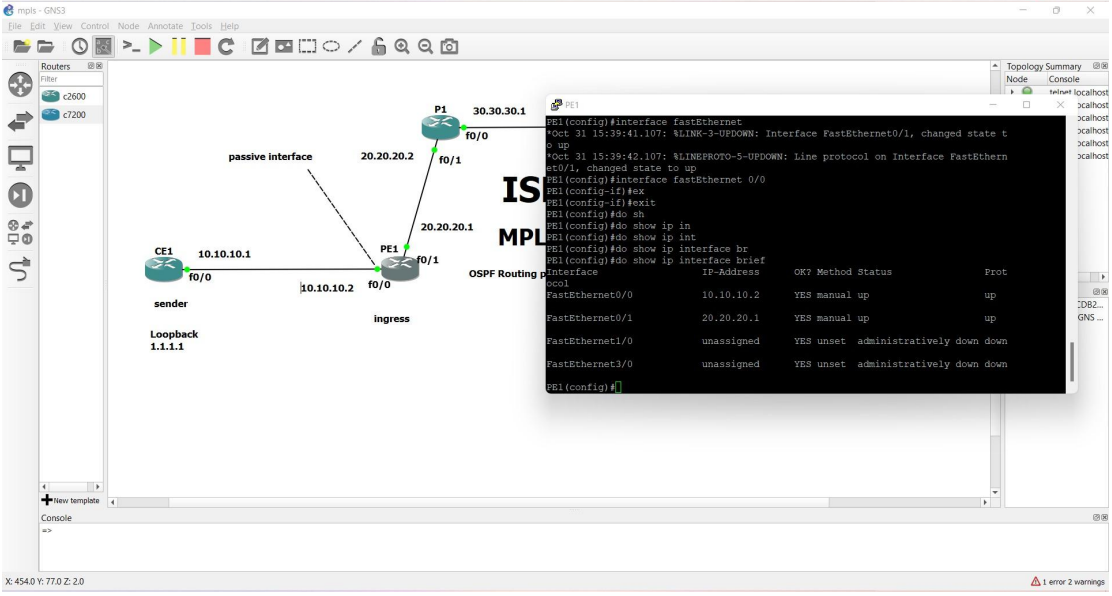
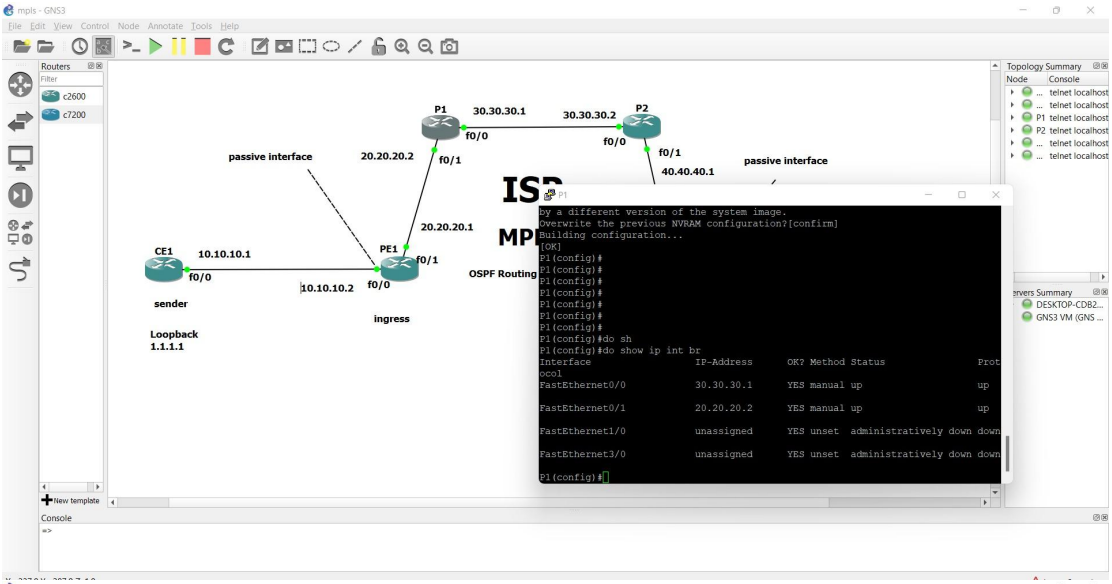
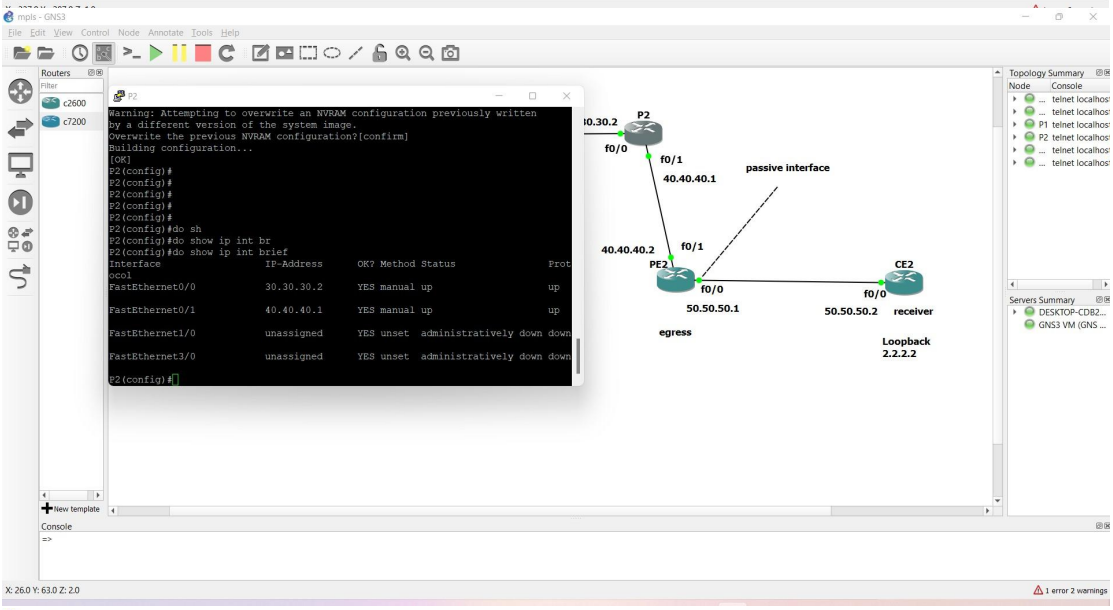
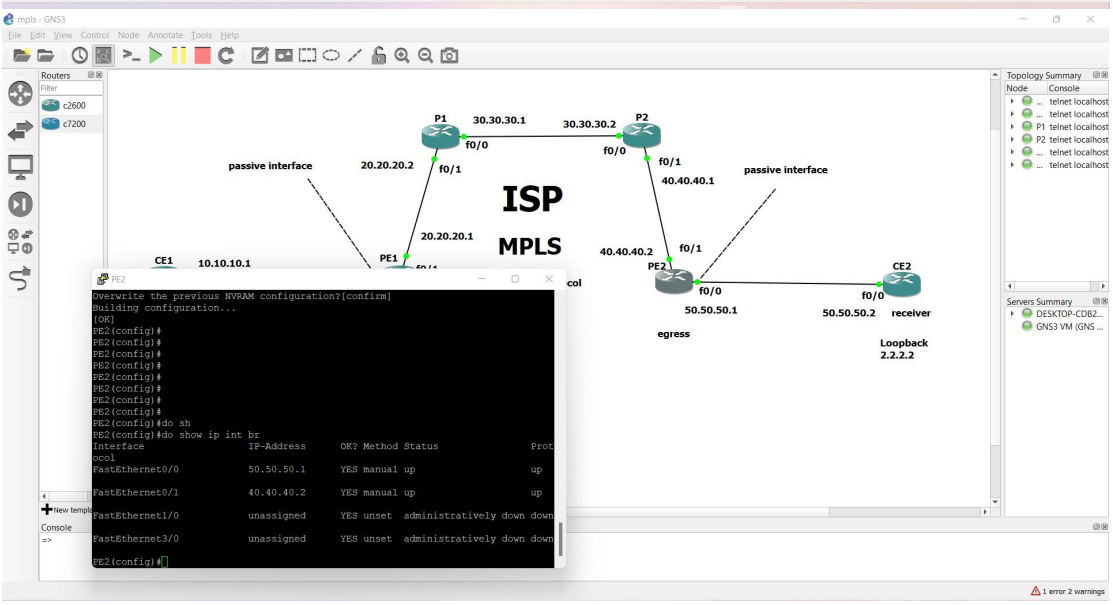

mpls - GNS3
File Edit View Control Node Annotate Tools Help
Routers
c2600
c7200
P1
30.30.30.1
30.30.30.2
P2
f0/0
f0/1
passive interface
20.20.20.2
ISP
MPLS
20.20.20.1
40.40.40.1
passive interface
CE1
10.10.10.1
PE1
OSPF Routing protocol
40.40.40.2
PE2
CE2
10.10.10.2
50.50.50.1
50.50.50.2
sender
ingress
egress
receiver
Loopback
1.1.1.1
Loopback
2.2.2.2
Topology Summary
Node Console
telnet localhost:5004
telnet localhost:5005
P1 telnet localhost:5001
P2 telnet localhost:5002
telnet localhost:5000
telnet localhost:5003
Servers Summary
DESKTOP-CDB2C45 CPU ...
GNS3 VM (GNS VM) CPU ...
New template
Console
=>
X: 276.0 Y: -174.0 Z: 1.0
1 error 2 warnings

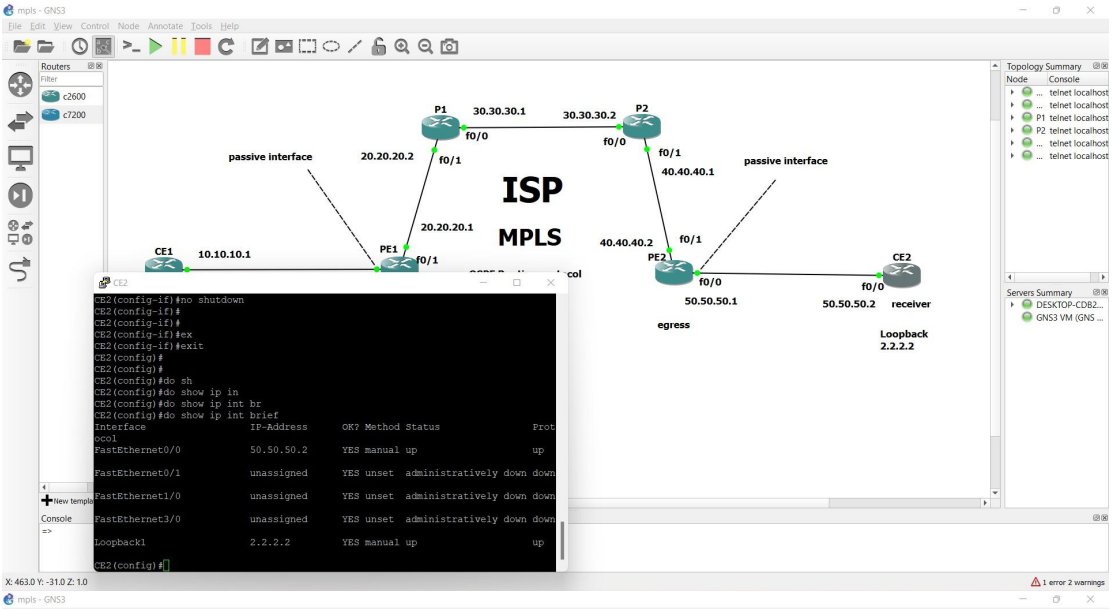

mpls - GNS3
File Edit View Control Node Annotate Tools Help
Routers
c2600
c7200
P1
30.30.30.1
30.30.30.2
P2
f0/0
passive interface
20.20.20.2
CE1
10.10.10.1
PE1
f0/1
10.10.10.2
sender
ingress
Loopback
1.1.1.1
CE1(config)#
CE1(config)#do show ip int brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.10.10.1 YES manual up up
FastEthernet0/1 unassigned YES unset administratively down down
FastEthernet1/0 unassigned YES unset administratively down down
FastEthernet3/0 unassigned YES unset administratively down down
Loopback1 1.1.1.1 YES manual up up
CE1(config)#
Topology Summary
Node Console
telnet localhost:5004
telnet localhost:5005
P1 telnet localhost:5001
P2 telnet localhost:5002
telnet localhost:5000
telnet localhost:5003
Servers Summary
DESKTOP-CDB2C45 CPU ...
GNS3 VM (GNS VM) CPU ...
New template
Console
=>
X: -551.0 Y: -2.0 Z: 2.0
1 error 2 warnings

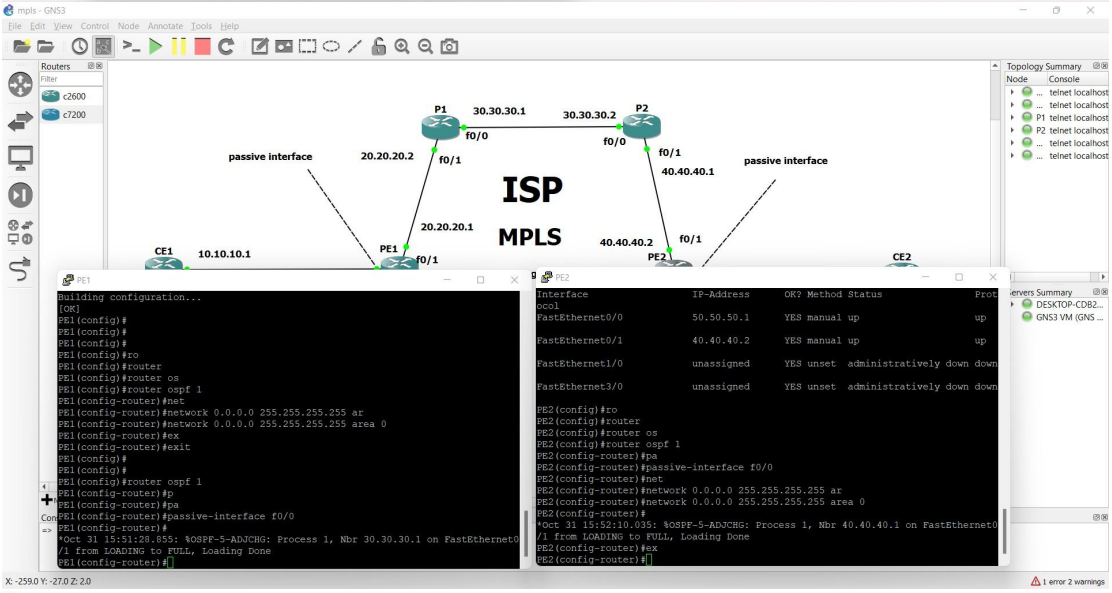

mpls - GNS3
File Edit View Control Node Annotate Tools Help
Routers
c2600
c7200
P1
30.30.30.1
f0/0
f0/1
passive interface
20.20.20.2
20.20.20.1
CE1
10.10.10.1
PE1
10.10.10.2
sender
ingress
Loopback
1.1.1.1
PE1(config)#interface fastEthernet
*Oct 31 15:39:41.107: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Oct 31 15:39:42.107: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
PE1(config)#interface fastEthernet 0/0
PE1(config-if)#ex
PE1(config-if)#exit
PE1(config)#do sh
PE1(config)#do show ip in
PE1(config)#do show ip int
PE1(config)#do show ip interface br
PE1(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.10.10.2 YES manual up up
FastEthernet0/1 20.20.20.1 YES manual up up
FastEthernet1/0 unassigned YES unset administratively down down
FastEthernet3/0 unassigned YES unset administratively down down
PE1(config)#
Topology Summary
Node Console
Servers Summary
New template
Console
=>
X: 454.0 Y: 77.0 Z: 2.0
1 error 2 warnings

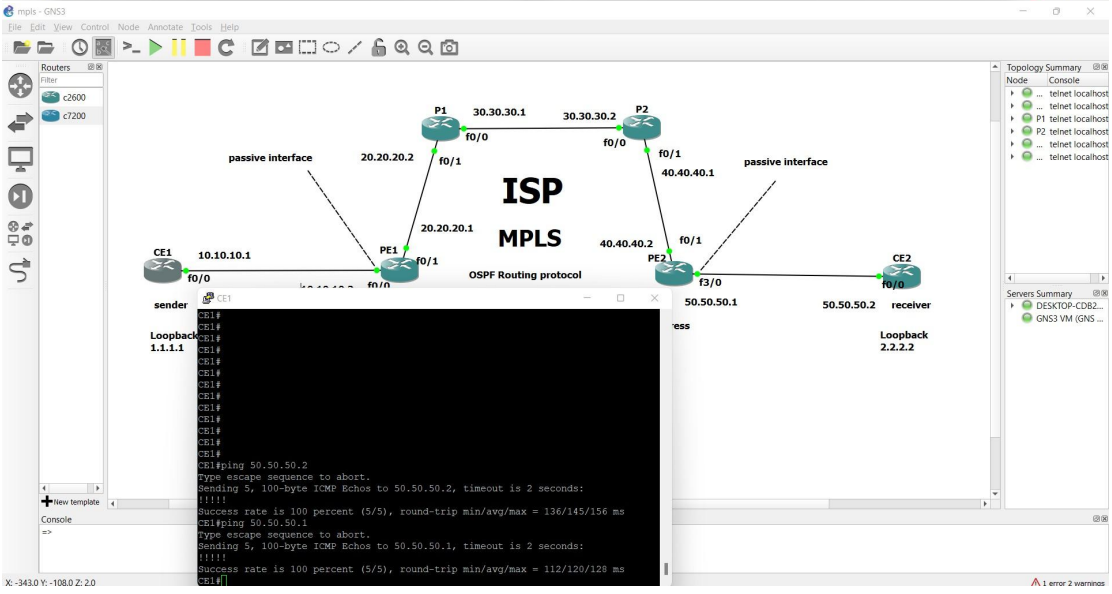

mpls - GNS3
P1
30.30.30.1
30.30.30.2
P2
f0/0
f0/1
passive interface
20.20.20.2
40.40.40.1
ISP
MPLS
20.20.20.1
CE1
10.10.10.1
PE1
OSPF Routing
10.10.10.2
sender
ingress
Loopback
1.1.1.1
P1(config)#do show ip int br
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 30.30.30.1 YES manual up up
FastEthernet0/1 20.20.20.2 YES manual up up
FastEthernet1/0 unassigned YES unset administratively down down
FastEthernet3/0 unassigned YES unset administratively down down

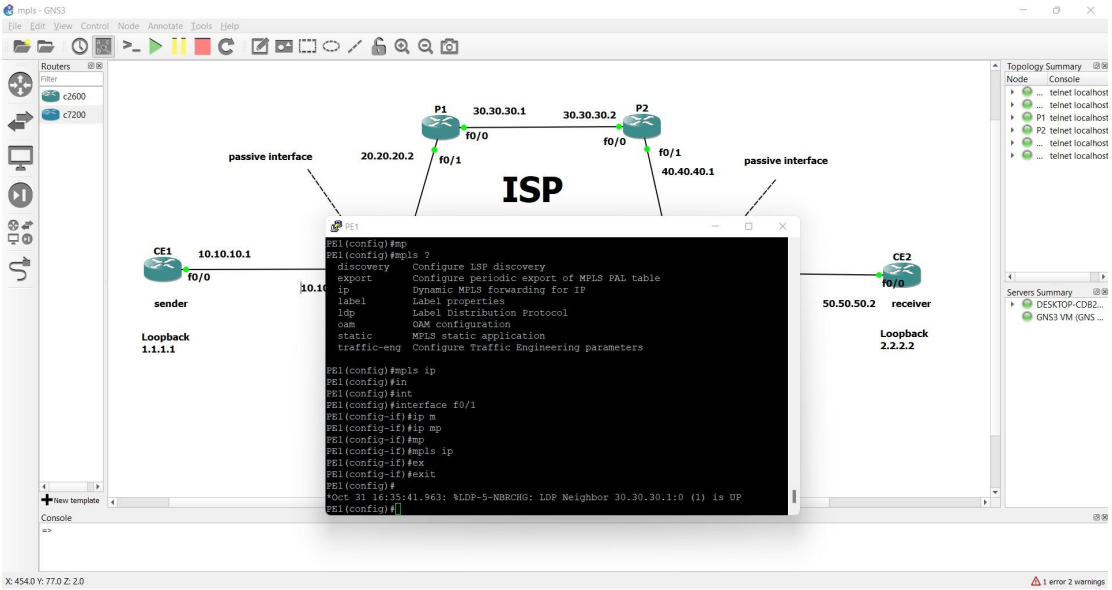

mpls - GNS3
P2
Warning: Attempting to overwrite an NVRAM configuration previously written by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
P2(config)#do show ip int br
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 30.30.30.2 YES manual up up
FastEthernet0/1 40.40.40.1 YES manual up up
FastEthernet1/0 unassigned YES unset administratively down down
FastEthernet3/0 unassigned YES unset administratively down down
P2
f0/0
f0/1
passive interface
40.40.40.1
40.40.40.2
PE2
CE2
50.50.50.1
50.50.50.2
receiver
egress
Loopback
2.2.2.2

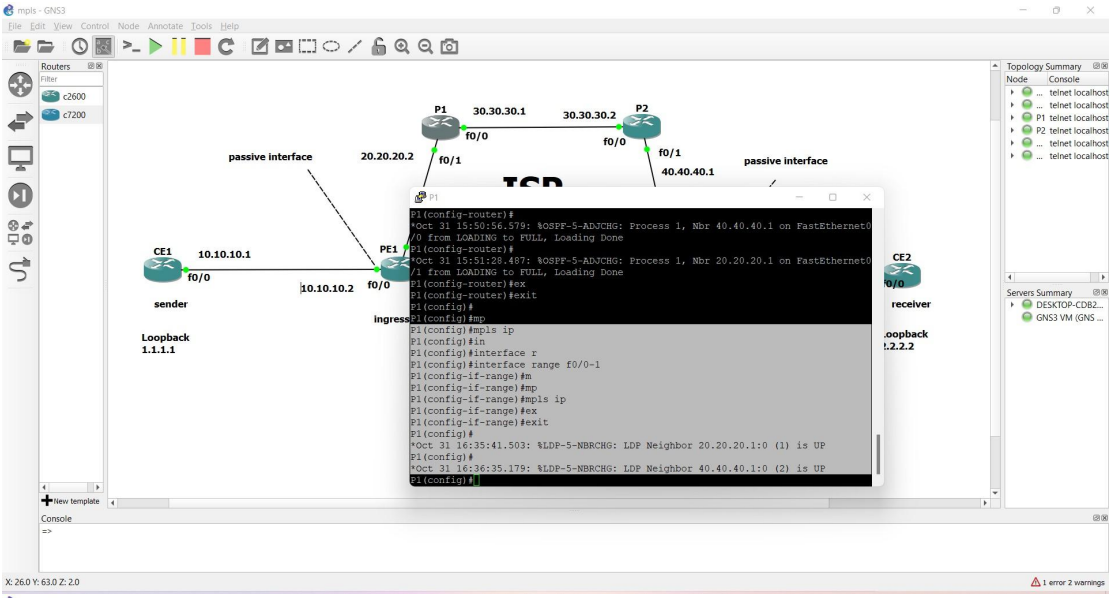

mpls - GNS3
PE2
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
PE2(config)#do show ip int br
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 50.50.50.1 YES manual up up
FastEthernet0/1 40.40.40.2 YES manual up up
FastEthernet1/0 unassigned YES unset administratively down down
FastEthernet3/0 unassigned YES unset administratively down down
P1
30.30.30.1
30.30.30.2
P2
f0/0
f0/1
passive interface
20.20.20.2
40.40.40.1
ISP
MPLS
20.20.20.1
CE1
10.10.10.1
PE1
40.40.40.2
PE2
CE2
50.50.50.1
50.50.50.2
receiver
egress
Loopback
2.2.2.2

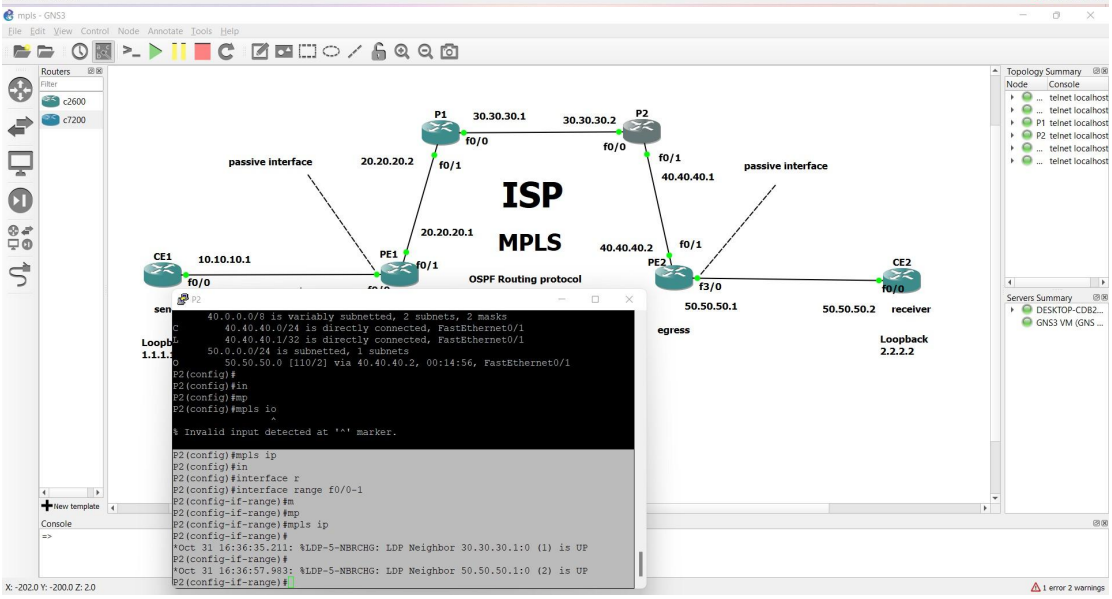

P1
30.30.30.1
30.30.30.2
P2
f0/0
f0/0
f0/1
f0/1
passive interface
20.20.20.2
40.40.40.1
passive interface
ISP
MPLS
20.20.20.1
40.40.40.2
f0/1
CE1
10.10.10.1
PE1
f0/1
PE2
CE2
f0/0
f0/0
50.50.50.1
50.50.50.2
receiver
egress
Loopback
2.2.2.2
CE2(config-if)#no shutdown
CE2(config-if)#
CE2(config-if)#
CE2(config-if)#ex
CE2(config-if)#exit
CE2(config)#
CE2(config)#
CE2(config)#do sh
CE2(config)#do show ip in
CE2(config)#do show ip int br
CE2(config)#do show ip int brief
Interface IP-Address OK? Method Status Prot
ocol
FastEthernet0/0 50.50.50.2 YES manual up up
FastEthernet0/1 unassigned YES unset administratively down down
FastEthernet1/0 unassigned YES unset administratively down down
FastEthernet3/0 unassigned YES unset administratively down down
Loopback1 2.2.2.2 YES manual up up
CE2(config)#

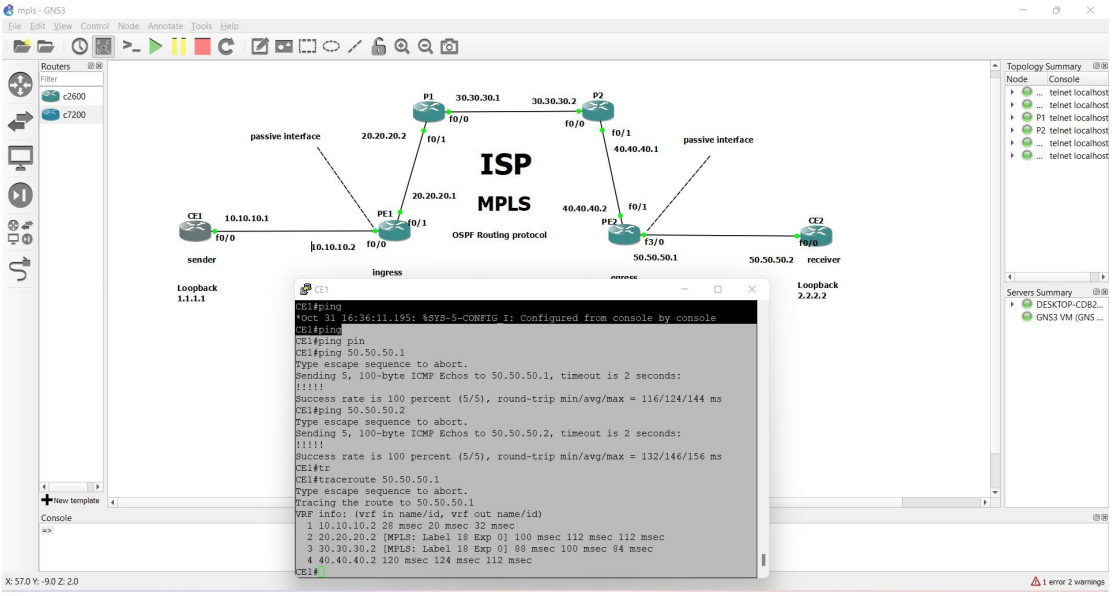

P1
30.30.30.1
30.30.30.2
P2
f0/0
f0/0
f0/1
f0/1
passive interface
20.20.20.2
40.40.40.1
passive interface
ISP
MPLS
20.20.20.1
40.40.40.2
f0/1
CE1
10.10.10.1
PE1
f0/1
PE2
CE2
Building configuration...
[OK]
PE1(config)#
PE1(config)#
PE1(config)#
PE1(config)#ro
PE1(config)#router
PE1(config)#router os
PE1(config)#router ospf 1
PE1(config-router)#net
PE1(config-router)#network 0.0.0.0 255.255.255.255 ar
PE1(config-router)#network 0.0.0.0 255.255.255.255 area 0
PE1(config-router)#ex
PE1(config-router)#exit
PE1(config)#
PE1(config)#
PE1(config)#router ospf 1
PE1(config-router)#p
PE1(config-router)#pa
PE1(config-router)#passive-interface f0/0
PE1(config-router)#
*Oct 31 15:51:28.855: %OSPF-5-ADJCHG: Process 1, Nbr 30.30.30.1 on FastEthernet0
/1 from LOADING to FULL, Loading Done
PE1(config-router)#
Interface IP-Address OK? Method Status Prot
ocol
FastEthernet0/0 50.50.50.1 YES manual up up
FastEthernet0/1 40.40.40.2 YES manual up up
FastEthernet1/0 unassigned YES unset administratively down down
FastEthernet3/0 unassigned YES unset administratively down down
PE2(config)#ro
PE2(config)#router
PE2(config)#router os
PE2(config)#router ospf 1
PE2(config-router)#pa
PE2(config-router)#passive-interface f0/0
PE2(config-router)#net
PE2(config-router)#network 0.0.0.0 255.255.255.255 ar
PE2(config-router)#network 0.0.0.0 255.255.255.255 area 0
PE2(config-router)#
*Oct 31 15:52:10.035: %OSPF-5-ADJCHG: Process 1, Nbr 40.40.40.1 on FastEthernet0
/1 from LOADING to FULL, Loading Done
PE2(config-router)#ex
PE2(config-router)#

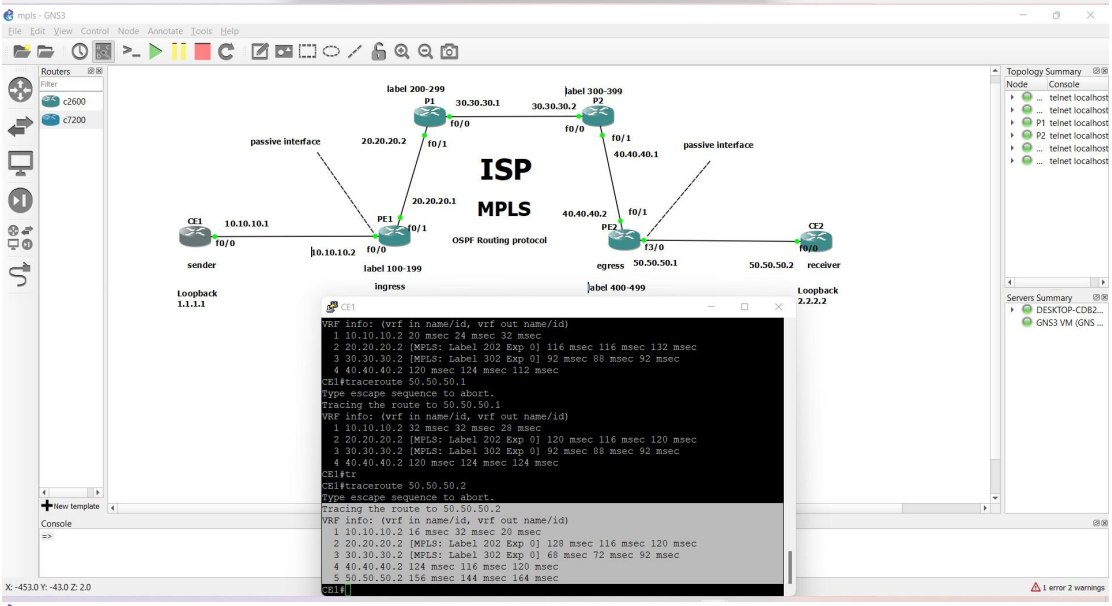

P1
30.30.30.1
30.30.30.2
P2
f0/0
f0/0
f0/1
f0/1
passive interface
20.20.20.2
40.40.40.1
passive interface
ISP
MPLS
20.20.20.1
40.40.40.2
f0/1
CE1
10.10.10.1
PE1
f0/1
PE2
CE2
f0/0
OSPF Routing protocol
f3/0
sender
Loopback
1.1.1.1
50.50.50.1
50.50.50.2
receiver
Loopback
2.2.2.2
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#
CE1#ping 50.50.50.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 50.50.50.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 136/145/156 ms
CE1#ping 50.50.50.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 50.50.50.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 112/120/128 ms
CE1#

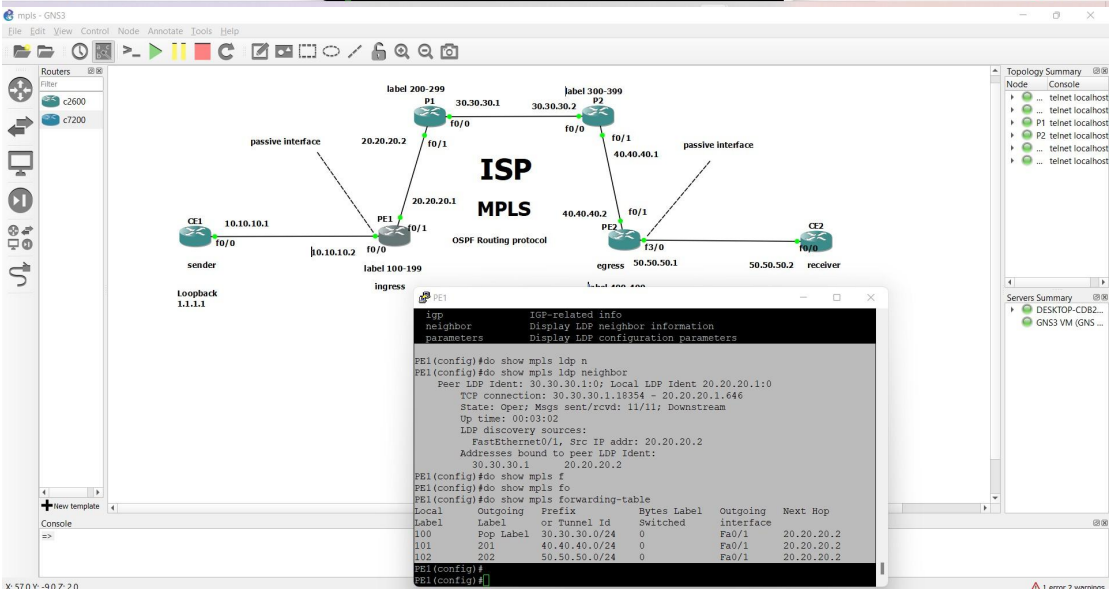

mpls - GNS3
P1
30.30.30.1
30.30.30.2
P2
f0/0
f0/0
f0/1
f0/1
20.20.20.2
40.40.40.1
passive interface
passive interface
ISP
CE1
10.10.10.1
f0/0
sender
Loopback
1.1.1.1
50.50.50.2
receiver
Loopback
2.2.2.2
CE2
f0/0
PE1
PE1(config)#mp
PE1(config)#mpls ?
discovery Configure LSP discovery
export Configure periodic export of MPLS PAL table
ip Dynamic MPLS forwarding for IP
label Label properties
ldp Label Distribution Protocol
oam OAM configuration
static MPLS static application
traffic-eng Configure Traffic Engineering parameters
PE1(config)#mpls ip
PE1(config)#in
PE1(config)#int
PE1(config)#interface f0/1
PE1(config-if)#ip m
PE1(config-if)#ip mp
PE1(config-if)#mp
PE1(config-if)#mpls ip
PE1(config-if)#ex
PE1(config-if)#exit
PE1(config)#
*Oct 31 16:35:41.963: %LDP-5-NBRCHG: LDP Neighbor 30.30.30.1:0 (1) is UP
PE1(config)#
Topology Summary
Servers Summary
Console
X: 454.0 Y: 77.0 Z: 2.0
1 error 2 warnings

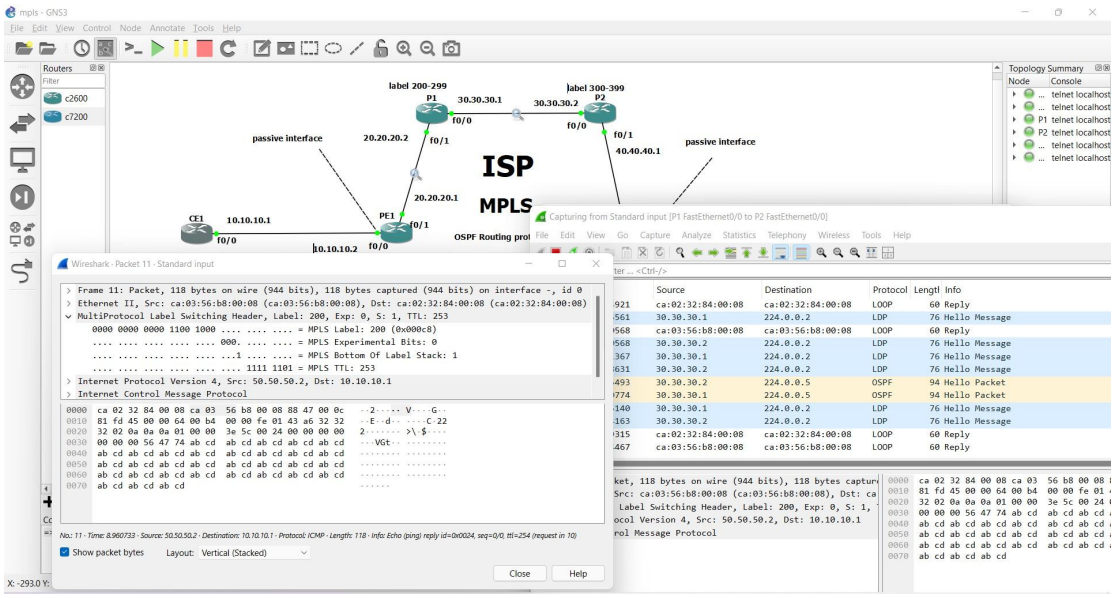

mpls - GNS3
P1
30.30.30.1
30.30.30.2
P2
f0/0
f0/0
f0/1
f0/1
20.20.20.2
40.40.40.1
passive interface
passive interface
CE1
10.10.10.1
f0/0
PE1
10.10.10.2
f0/0
sender
ingress
CE2
f0/0
receiver
Loopback
1.1.1.1
P1(config-router)#
*Oct 31 15:50:56.579: %OSPF-5-ADJCHG: Process 1, Nbr 40.40.40.1 on FastEthernet0
/0 from LOADING to FULL, Loading Done
P1(config-router)#
*Oct 31 15:51:28.487: %OSPF-5-ADJCHG: Process 1, Nbr 20.20.20.1 on FastEthernet0
/1 from LOADING to FULL, Loading Done
P1(config-router)#ex
P1(config-router)#exit
P1(config)#
P1(config)#mp
P1(config)#mpls ip
P1(config)#in
P1(config)#interface r
P1(config)#interface range f0/0-1
P1(config-if-range)#m
P1(config-if-range)#mp
P1(config-if-range)#mpls ip
P1(config-if-range)#ex
P1(config-if-range)#exit
P1(config)#
*Oct 31 16:35:41.503: %LDP-5-NBRCHG: LDP Neighbor 20.20.20.1:0 (1) is UP
P1(config)#
*Oct 31 16:36:35.179: %LDP-5-NBRCHG: LDP Neighbor 40.40.40.1:0 (2) is UP
P1(config)#
Topology Summary
Servers Summary
Console
X: 26.0 Y: 63.0 Z: 2.0
1 error 2 warnings


mpls - GNS3
P1
30.30.30.1
30.30.30.2
P2
f0/0
f0/0
f0/1
f0/1
20.20.20.2
40.40.40.1
passive interface
passive interface
ISP
MPLS
20.20.20.1
40.40.40.2
CE1
10.10.10.1
f0/0
PE1
f0/1
PE2
f0/1
f3/0
50.50.50.1
OSPF Routing protocol
egress
50.50.50.2
receiver
CE2
f0/0
Loopback
2.2.2.2
P2
40.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 40.40.40.0/24 is directly connected, FastEthernet0/1
L 40.40.40.1/32 is directly connected, FastEthernet0/1
50.0.0.0/24 is subnetted, 1 subnets
O 50.50.50.0 [110/2] via 40.40.40.2, 00:14:56, FastEthernet0/1
P2(config)#
P2(config)#in
P2(config)#mp
P2(config)#mpls io
^
% Invalid input detected at '^' marker.
P2(config)#mpls ip
P2(config)#in
P2(config)#interface r
P2(config)#interface range f0/0-1
P2(config-if-range)#m
P2(config-if-range)#mp
P2(config-if-range)#mpls ip
P2(config-if-range)#
*Oct 31 16:36:35.211: %LDP-5-NBRCHG: LDP Neighbor 30.30.30.1:0 (1) is UP
P2(config-if-range)#
*Oct 31 16:36:57.983: %LDP-5-NBRCHG: LDP Neighbor 50.50.50.1:0 (2) is UP
P2(config-if-range)#
Topology Summary
Servers Summary
Console
X: -202.0 Y: -200.0 Z: 2.0
1 error 2 warnings


mpls - GNS3
ISP
MPLS
OSPF Routing protocol
P1
P2
PE1
PE2
CE1
CE2
30.30.30.1
30.30.30.2
20.20.20.2
20.20.20.1
40.40.40.1
40.40.40.2
10.10.10.1
10.10.10.2
50.50.50.1
50.50.50.2
passive interface
sender
receiver
ingress
egress
Loopback 1.1.1.1
Loopback 2.2.2.2
CE1#ping
*Oct 31 16:36:11.195: %SYS-5-CONFIG_I: Configured from console by console
CE1#ping
CE1#ping pin
CE1#ping 50.50.50.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 50.50.50.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 116/124/144 ms
CE1#ping 50.50.50.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 50.50.50.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 132/146/156 ms
CE1#tr
CE1#traceroute 50.50.50.1
Type escape sequence to abort.
Tracing the route to 50.50.50.1
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.10.2 28 msec 20 msec 32 msec
2 20.20.20.2 [MPLS: Label 18 Exp 0] 100 msec 112 msec 112 msec
3 30.30.30.2 [MPLS: Label 18 Exp 0] 88 msec 100 msec 84 msec
4 40.40.40.2 120 msec 124 msec 112 msec
CE1#


mpls - GNS3
label 200-299
label 300-399
label 100-199
label 400-499
ISP
MPLS
OSPF Routing protocol
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.10.2 20 msec 24 msec 32 msec
2 20.20.20.2 [MPLS: Label 202 Exp 0] 116 msec 116 msec 132 msec
3 30.30.30.2 [MPLS: Label 302 Exp 0] 92 msec 88 msec 92 msec
4 40.40.40.2 120 msec 124 msec 112 msec
CE1#traceroute 50.50.50.1
Type escape sequence to abort.
Tracing the route to 50.50.50.1
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.10.2 32 msec 32 msec 28 msec
2 20.20.20.2 [MPLS: Label 202 Exp 0] 120 msec 116 msec 120 msec
3 30.30.30.2 [MPLS: Label 302 Exp 0] 92 msec 88 msec 92 msec
4 40.40.40.2 120 msec 124 msec 124 msec
CE1#tr
CE1#traceroute 50.50.50.2
Type escape sequence to abort.
Tracing the route to 50.50.50.2
VRF info: (vrf in name/id, vrf out name/id)
1 10.10.10.2 16 msec 32 msec 20 msec
2 20.20.20.2 [MPLS: Label 202 Exp 0] 128 msec 116 msec 120 msec
3 30.30.30.2 [MPLS: Label 302 Exp 0] 68 msec 72 msec 92 msec
4 40.40.40.2 124 msec 116 msec 120 msec
5 50.50.50.2 156 msec 144 msec 164 msec
CE1#


mpls - GNS3
label 200-299
label 300-399
label 100-199
ISP
MPLS
OSPF Routing protocol
igp IGP-related info
neighbor Display LDP neighbor information
parameters Display LDP configuration parameters
PE1(config)#do show mpls ldp n
PE1(config)#do show mpls ldp neighbor
Peer LDP Ident: 30.30.30.1:0; Local LDP Ident 20.20.20.1:0
TCP connection: 30.30.30.1.18354 - 20.20.20.1.646
State: Oper; Msgs sent/rcvd: 11/11; Downstream
Up time: 00:03:02
LDP discovery sources:
FastEthernet0/1, Src IP addr: 20.20.20.2
Addresses bound to peer LDP Ident:
30.30.30.1 20.20.20.2
PE1(config)#do show mpls f
PE1(config)#do show mpls fo
PE1(config)#do show mpls forwarding-table
Local Label Outgoing Label Prefix or Tunnel Id Bytes Label Switched Outgoing interface Next Hop
100 Pop Label 30.30.30.0/24 0 Fa0/1 20.20.20.2
101 201 40.40.40.0/24 0 Fa0/1 20.20.20.2
102 202 50.50.50.0/24 0 Fa0/1 20.20.20.2
PE1(config)#
PE1(config)#


mpls - GNS3
File Edit View Control Node Annotate Tools Help
Routers
Filter
c2600
c7200
label 200-299
P1
30.30.30.1
f0/0
label 300-399
P2
30.30.30.2
f0/0
f0/1
40.40.40.1
passive interface
20.20.20.2
f0/1
passive interface
ISP
MPLS
20.20.20.1
CE1
10.10.10.1
f0/0
PE1
f0/1
10.10.10.2
f0/0
OSPF Routing prot
Topology Summary
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Console
telnet localhost
P1
P2
Capturing from Standard input [P1 FastEthernet0/0 to P2 FastEthernet0/0]
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Source
Destination
Protocol
Lengtl
Info
ca:02:32:84:00:08
ca:02:32:84:00:08
LOOP
60 Reply
30.30.30.1
224.0.0.2
LDP
76 Hello Message
ca:03:56:b8:00:08
ca:03:56:b8:00:08
LOOP
60 Reply
30.30.30.2
224.0.0.2
LDP
76 Hello Message
30.30.30.1
224.0.0.2
LDP
76 Hello Message
30.30.30.2
224.0.0.2
LDP
76 Hello Message
30.30.30.2
224.0.0.5
OSPF
94 Hello Packet
30.30.30.1
224.0.0.5
OSPF
94 Hello Packet
30.30.30.1
224.0.0.2
LDP
76 Hello Message
30.30.30.2
224.0.0.2
LDP
76 Hello Message
ca:02:32:84:00:08
ca:02:32:84:00:08
LOOP
60 Reply
ca:03:56:b8:00:08
ca:03:56:b8:00:08
LOOP
60 Reply
Wireshark · Packet 11 · Standard input
Frame 11: Packet, 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface -, id 0
Ethernet II, Src: ca:03:56:b8:00:08 (ca:03:56:b8:00:08), Dst: ca:02:32:84:00:08 (ca:02:32:84:00:08)
MultiProtocol Label Switching Header, Label: 200, Exp: 0, S: 1, TTL: 253
0000 0000 0000 1100 1000 .... = MPLS Label: 200 (0x000c8)
.... 000. .... = MPLS Experimental Bits: 0
....1 .... = MPLS Bottom Of Label Stack: 1
.... 1111 1101 = MPLS TTL: 253
Internet Protocol Version 4, Src: 50.50.50.2, Dst: 10.10.10.1
Internet Control Message Protocol
0000 ca 02 32 84 00 08 ca 03 56 b8 00 08 88 47 00 0c
0010 81 fd 45 00 00 64 00 b4 00 00 fe 01 43 a6 32 32
0020 32 02 0a 0a 0a 01 00 00 3e 5c 00 24 00 00 00 00
0030 00 00 00 56 47 74 ab cd ab cd ab cd ab cd ab cd
0040 ab cd ab cd ab cd ab cd ab cd ab cd ab cd ab cd
0050 ab cd ab cd ab cd ab cd ab cd ab cd ab cd ab cd
0060 ab cd ab cd ab cd ab cd ab cd ab cd ab cd ab cd
0070 ab cd ab cd ab cd
No.: 11 · Time: 8.960733 · Source: 50.50.50.2 · Destination: 10.10.10.1 · Protocol: ICMP · Length: 118 · Info: Echo (ping) reply id=0x0024, seq=0/0, ttl=254 (request in 10)
Show packet bytes
Layout: Vertical (Stacked)
Close
Help
X: -293.0 Y: